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dp

Ques: Longest Common Substring (LCS lekin vo nahi)

deep preet ee

raghav
madhav

3 len

raghav
madhav
1+
ragha
madha
1+
ragh
madh
1+
rag
mad
1

return 0

this '1' gets added to 3

LCS = ahav subsequence
LCS = hav (substring)

raghav
aditya

ragha
aditya

raghav
adity

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Iterative DP for Longest common substring

preet , deep

	d	e	e	p
p	0	0	0	1
r	0	0	0	0
e	0	1	1	0
e	0	1	2	0
t	0	0	0	0

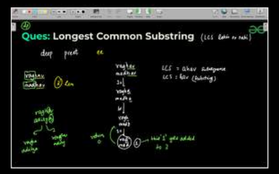
dp

dp[i][j] means preet(0,i) & deep(0,j)

take ki max common substring length finishing at preet(i) & deep(j)

pre
de

dee
pree



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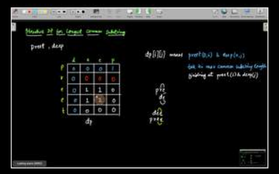
anuj ananya → an

	a	n	a	n	y	a
a	1	0	1	0	0	1
n	0	2	0	2	0	0
u	0	0	0	0	0	0
j	0	0	0	0	0	0



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Count no. of palindromic substring

Ques: Longest Palindromic Substring

$\neq LCS(S, rev(S))$

LPS = 1

S = raghavgarg

S' = gragvahgar

common → 3 ✗

i to j palindrome exists if str[i] == str[j] & i+1 to j-1 tak palindrome hai.

a n a n y a

if this is palindrome len + 2

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anonya

	0	1	2	3	4	5
0	1	0	3	0	0	0
1	X	1	0	3	0	0
2	X	X	1	0	0	0
3	X	X	X	1	0	0
4	X	X	X	X	1	0
5	X	X	X	X	X	1

dp

0 1 2 3 4 5

a n a n y a

0,0 1,1 2,2 3,3 4,4 5,5

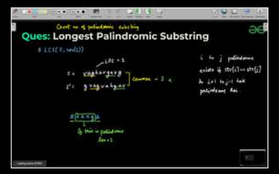
0,1 1,2 2,3 , 3,4, 4,5

0,2 1,3, 2,4, 3,5

0,3 1,4 , 2,5

0,4 1,5

0,5



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